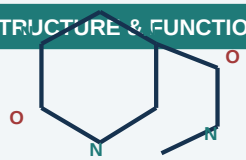
CAFFEINE

IDENTITY

C8H10N4O2 • 194.19 g/mol

IUPAC name: 1,3,7-trimethylpurine-2,6-dione

STRUCTURE & FUNCTIONAL GROUPS



Schematic only; use a verified structure drawing for exact bonds.

- Fused nitrogen-containing ring system
- Two carbonyl groups (C=O)
- Three methyl groups attached to nitrogen

KEY GROUPS

Methylxanthine; carbonyls; ring nitrogens

IMPORTANT PROPERTIES

- White crystalline solid at room temperature.
- Water solubility increases with temperature.
- Contains several hydrogen-bond acceptor sites, but no O–H or N–H donor group.

WHAT IS AMAZING ABOUT IT?

It competes with a sleep-related signal

Caffeine can promote alertness by blocking adenosine receptors. Adenosine signaling normally contributes to the feeling of sleep pressure; caffeine does not replace sleep.

USES & VALUE

- Coffee, tea, and other beverages
- Ingredient in some medicines and consumer products
- Used in decaffeination research and extraction studies

RISKS & SAFETY

- Too much may cause restlessness, anxiety, sleep disruption, or palpitations.
- Sensitivity varies; intake should be considered alongside age, pregnancy, medicines, and health factors.

CHEMICAL ENGINEERING CONNECTION

Solid–liquid extraction, adsorption, crystallization, and quality control are relevant to caffeine recovery and decaffeinated-product manufacture.

SELECTED REFERENCES

National Center for Biotechnology Information. (n.d.). PubChem compound summary for caffeine (CID 2519). https://pubchem.ncbi.nlm.nih.gov/compound/2519
European Food Safety Authority. (2015). Scientific opinion on the safety of caffeine. EFSA Journal, 13(5), 4102. https://doi.org/10.2903/j.efsa.2015.4102